

CAPGEMINI HACKATHON

Maestro AI

Multi-Agent Music Composition System

From mood to music — in seconds. Powered by 6 specialized AI agents.

6

AI Agents

3

Output Versions

<8s

Generation Time

60%

LLM Cost Savings

Musicians Lack Tools That Think

The creative process is slow, expensive, and disconnected from real musical patterns.



Hours of Setup

Musicians spend 2–4 hrs arranging before one note is composed.



High Studio Cost

\$200–\$500/hr studio rates block emerging artists from experimenting.



Generic AI Output

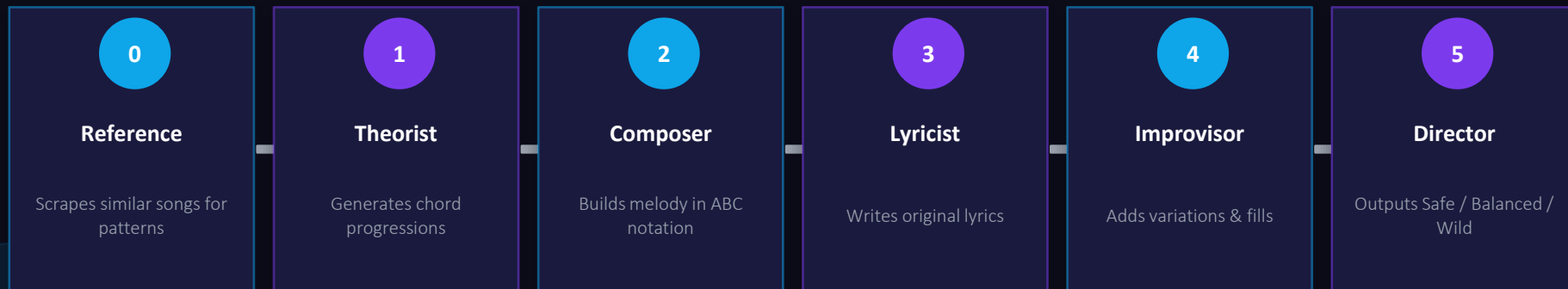
Existing tools produce musically incorrect, style-deaf compositions.



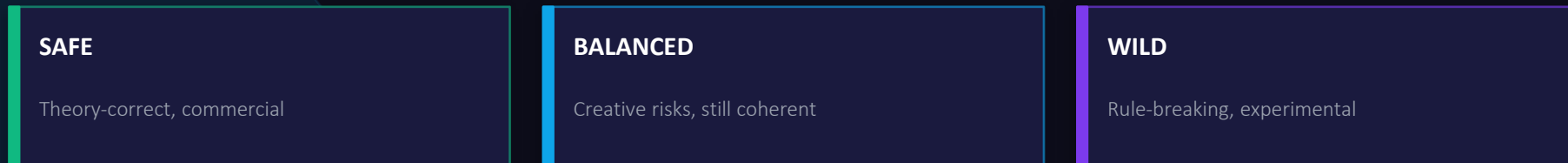
No Creative Context

No tool learns from real songs to inform what actually sounds good.

Maestro AI — 6-Agent Pipeline



FINAL OUTPUT — 3 CREATIVE TIERS



Industrial-Grade by Design

Frontend	React 18 + TypeScript + Tone.js
API Gateway	FastAPI + JWT Auth + Rate Limiting
Agent Layer	LangChain Supervisor · 6 Agents
LLM Router	LiteLLM — Claude + GPT-4o fallback
Cache / Queue	Redis (semantic cache) + Celery
Data	PostgreSQL · S3 · Prometheus + Grafana

60%

LLM Cost Cut

Semantic caching in Redis

<8s

End-to-End

Parallel Lyricist+Improvisor

3×

Model Fallback

Claude → GPT auto-switch

∞

Scalable

Kubernetes HPA autoscale

Built to Scale from Day One

Free	Pro	Studio	Enterprise
₹0/mo	₹499/mo	₹2499/mo	Custom
✓ 5 compositions/day ✓ 3 export formats ✓ Tone.js playback	✓ Unlimited compose ✓ MIDI export ✓ Genre blending	✓ Team workspace ✓ API access ✓ Custom agent tuning	✓ White-label ✓ On-prem deploy ✓ SLA + support

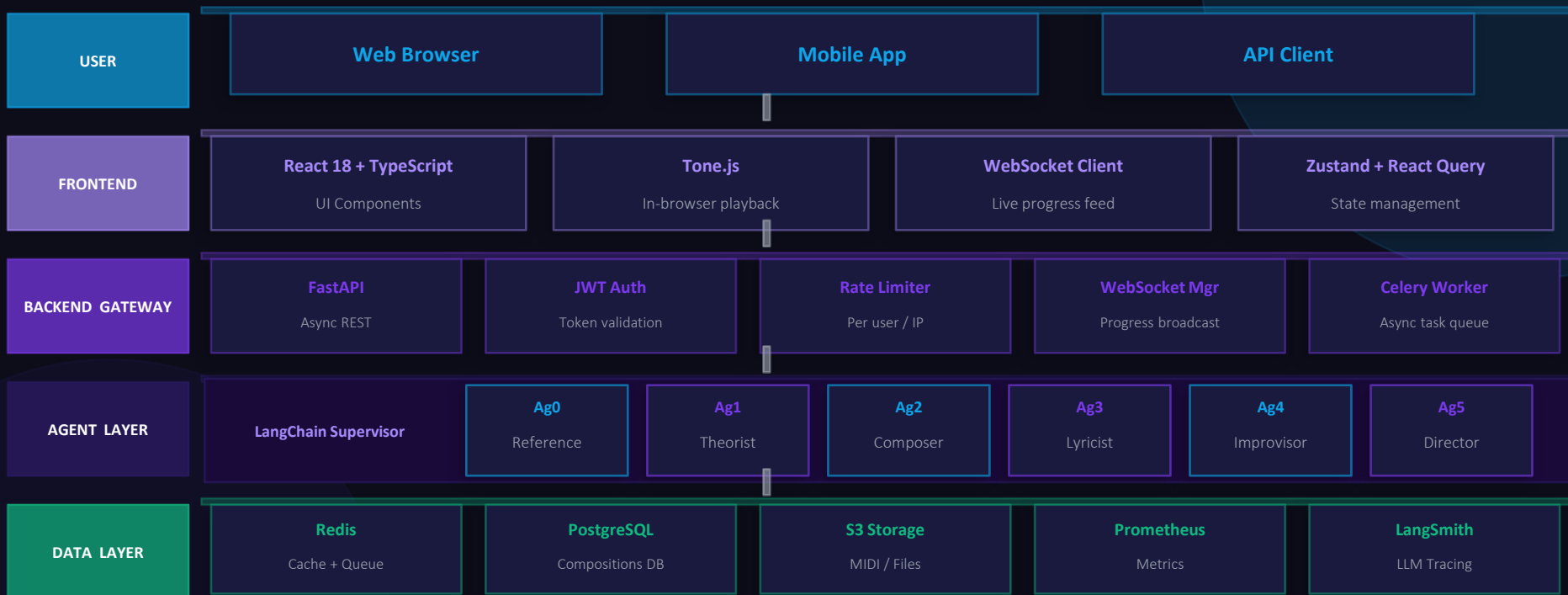
MARKET OPPORTUNITY

\$6.8B Music Tech Market 2024

11M+ Independent Musicians Globally

43% YoY AI Music Tool Growth

How Maestro AI Works End-to-End



Where the Market Falls Short — and We Don't

⚠ Existing AI Music Tools (Market Today)

VS

✓ Maestro AI

	Single LLM prompt → one output. No structured musical thinking or agent specialization.	How it Composes	6 specialist agents in sequence. Each one thinks deeply about its domain before passing context forward.
	Trained on general data. Cannot reference what actual hit songs in that genre actually sound like today.	Real Song Awareness	Agent 0 actively scrapes reference songs at runtime. Patterns are grounded in real, current music.
	One output per generation. To get variation, users manually re-prompt and get random results.	Output Variety	3 creative tiers every time — Safe, Balanced, Wild — each artistically intentional, not random.
	Black box. Users cannot lock a chord, adjust improvisation depth, or blend two genre styles together.	Creative Control	Full control: lock any element, slide improv level 1–10, blend genres proportionally (e.g. Jazz 60% + HipHop 40%).
	Every request hits the LLM API cold. Repeated style requests waste compute and money at scale.	Cost & Scalability	Redis semantic cache cuts LLM calls by ~60%. Kubernetes autoscaling handles 10,000+ concurrent users.

Maestro AI is not competing with audio synthesis tools — it's a composer's intelligent drafting partner, grounded in real musical knowledge.

The Future of Music Is Composed by Intelligence



Maestro AI turns a mood, a genre, and a key into a full composition
— chords, melody, and lyrics — in under 8 seconds.

Multi-Agent LangChain Pipeline

Real Song Reference Scraping

Creative Freedom + Improvisation

Industrial Deployment Ready